

AI & Machine Learning – Comprehensive Syllabus

Course Overview

This course introduces Artificial Intelligence (AI) and Machine Learning (ML) from fundamentals to practical implementation. Students will learn data preprocessing, supervised and unsupervised learning, model evaluation, neural networks, and real-world AI applications using Python and industry-standard libraries.

Duration: 12–16 Weeks (70–100 Hours)

Prerequisites:

- Basic Python Programming
 - Basic Mathematics (Algebra, Statistics)
 - Basic understanding of data handling
-

Module 1: Introduction to AI & Machine Learning (Week 1)

Topics

- What is Artificial Intelligence?
- Machine Learning vs AI vs Deep Learning
- Types of Machine Learning
- Applications of AI & ML
- AI Workflow
- Introduction to Python for ML

Practical Exercises

- Setting up Python ML environment
 - Basic data exploration using Python
-

Module 2: Python for Data Science (Week 2)

Topics

- NumPy Basics
- Pandas Basics
- DataFrames and Series
- Data Cleaning
- Handling Missing Values
- Data Visualization Basics (Matplotlib, Seaborn)

Practical Exercises

- Dataset loading and cleaning
 - Basic statistical analysis
 - Simple data visualization
-

Module 3: Mathematics for Machine Learning (Week 3)

Topics

- Linear Algebra Basics
- Probability Concepts
- Statistics (Mean, Median, Mode)
- Variance & Standard Deviation
- Correlation
- Gradient Concept (Intro)

Practical Exercises

- Statistical analysis using Python
 - Correlation matrix on dataset
-

Module 4: Data Preprocessing (Week 4)

Topics

- Data Collection
- Data Cleaning
- Handling Outliers
- Feature Scaling (Normalization, Standardization)
- Encoding Categorical Data
- Train-Test Split

Practical Exercises

- Titanic dataset preprocessing
 - Feature scaling exercises
-

Module 5: Supervised Learning – Regression (Week 5)

Topics

- Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Evaluation Metrics (MAE, MSE, RMSE, R^2)

Practical Exercises

- House price prediction
 - Salary prediction model
-

Module 6: Supervised Learning – Classification (Week 6)

Topics

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Trees
- Random Forest

- Naive Bayes

Practical Exercises

- Spam detection system
 - Iris flower classification
 - Customer churn prediction
-

Module 7: Model Evaluation & Optimization (Week 7)

Topics

- Confusion Matrix
- Precision, Recall, F1 Score
- ROC Curve
- Cross Validation
- Overfitting & Underfitting
- Hyperparameter Tuning

Practical Exercises

- Model comparison experiments
 - Accuracy improvement tasks
-

Module 8: Unsupervised Learning (Week 8)

Topics

- Clustering Concepts
- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Dimensionality Reduction (PCA)

Practical Exercises

- Customer segmentation
 - Market basket clustering
 - PCA visualization
-

Module 9: Feature Engineering (Week 9)

Topics

- Feature Selection
- Feature Importance
- Encoding Techniques
- Dimensionality Reduction
- Handling Imbalanced Data

Practical Exercises

- Feature optimization on dataset
 - Improving model accuracy
-

Module 10: Introduction to Deep Learning (Week 10)

Topics

- Neural Networks Basics
- Perceptron
- Activation Functions
- Forward & Backpropagation
- Introduction to TensorFlow/Keras

Practical Exercises

- Simple neural network model
 - Digit recognition (MNIST basic model)
-

Module 11: Natural Language Processing (NLP) (Week 11)

Topics

- Text Processing
- Tokenization
- Stop Words Removal
- Bag of Words
- TF-IDF
- Sentiment Analysis

Practical Exercises

- Sentiment classifier
 - Text classification model
-

Module 12: Model Deployment Basics (Week 12)

Topics

- Saving Models (Pickle/Joblib)
- Flask API for ML Models
- Introduction to FastAPI
- Model Deployment Concepts
- Introduction to Cloud Deployment

Practical Exercises

- Deploy ML model as API
 - Simple prediction web app
-

Module 13: Mini Project (Week 13)

Choose one:

- Spam Email Classifier

- Movie Recommendation System
 - Customer Segmentation System
 - House Price Prediction System
 - Sentiment Analysis Tool
 - Disease Prediction System
-

Module 14–16: Capstone Project

End-to-End AI/ML System Development

- Problem Definition
 - Data Collection
 - Data Preprocessing
 - Feature Engineering
 - Model Selection
 - Training & Evaluation
 - Hyperparameter Tuning
 - Deployment (Web/API)
 - Documentation
 - Final Presentation
-

Tools & Libraries

- Python
 - NumPy
 - Pandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - TensorFlow / Keras
 - PyTorch (optional)
 - Flask / FastAPI
 - Jupyter Notebook
 - Google Colab
-

Core Concepts Covered

- Artificial Intelligence Fundamentals
 - Machine Learning Algorithms
 - Data Preprocessing Techniques
 - Supervised & Unsupervised Learning
 - Model Evaluation & Optimization
 - Feature Engineering
 - Neural Networks & Deep Learning Basics
 - Natural Language Processing (NLP)
 - Model Deployment
-

Assessment Scheme

Component	Weightage
Assignments	15%
Lab Exercises	25%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of this course, learners will be able to:

- Understand core concepts of Artificial Intelligence and Machine Learning.
- Build and train supervised and unsupervised ML models.
- Perform data preprocessing and feature engineering effectively.
- Evaluate and optimize machine learning models using performance metrics.
- Develop basic neural networks for deep learning tasks.
- Perform text analysis using Natural Language Processing techniques.
- Build end-to-end machine learning pipelines.
- Deploy ML models using APIs and basic web frameworks.
- Work on real-world datasets and solve practical AI problems.
- Develop industry-ready AI/ML solutions for business and research applications.